The University of Azad Jammu and Kashmir Muzaffarabad

Task 01

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Course:	Operating System
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1. Chosen Domain

I chose the university registration domain. Every student knows how it works. Students pick courses, courses have classes, and classes meet in rooms. This makes it easy to find the main objects.

2. Candidate Conceptual Classes

Class	Why it is needed
Student	A person who signs up for courses.
Instructor	A teacher who teaches sections and gives grades.
Department	A group, like Computer Science, that owns courses.
Program	The degree a student is studying for.
Course	A subject that is taught, such as “Databases”.
Section	One class of a course in one term, with a time and a room.
Semester	The term in which classes run, with its deadlines.
Enrollment	Connects one student to one section and stores the grade.
Classroom	The room where a section meets.
Prerequisite	A course that must be passed before another course.

3. Attributes and Methods of Three Classes

Class	Attributes	Methods
Student	Student Id, name, email, program, credits Completed	register(section), drop(section), viewTimetable()
Course	courseCode, title, creditHours, department	addPrerequisite(course), getSections(semester)
Enrollment	student, section, status, grade, enrollDate	confirm(), withdraw(), assignGrade(grade)

4. Abstraction Decisions

I kept only the things needed to register for a course. I left out a student’s home address, photo and health details, because they do not help with registration. I also left out fee payments, library records and the full grading scale, because they belong to other systems. I kept grade as an attribute of Enrollment instead of making a new class. I made Course and Section separate, because a course is the idea and a section is one real class of it. This keeps the model small and clear.

5. Invented Software Object: Waitlist Manager

Waitlist Manager is not a real-world object, but it would be useful in software. When a section is full, it keeps a fair line of waiting students. If a student drops the section, it offers the free seat to the first student in line. It first checks the prerequisites and the timetable. If that student does not accept in time, it offers the seat to the next one. This saves time for the registrar’s office.